Byte Bistro Recipe Database

IS 202 – Deliverable 3: Database Report

Group 1 | November 1, 2026

1. Project Overview

Byte Bistro is our group's recipe database for small restaurants. It will help store recipes, ingredients, and cooking instructions in one place.

For this assignment, our group created SQL scripts for the database tables and sample data. We also prepared a Docker Compose file that uses PostgreSQL 15 to run the database in a container.

2. Database Tables

Our `CREATE_TABLES.sql` file defines six tables:

Table	Primary Key	Purpose
users	**user_id**	**Stores user names, emails, and roles.**
categories	**category_id**	**Stores recipe categories.**
recipes	**recipe_id**	**Stores recipes and their basic information.**
ingredients	**ingredient_id**	**Stores ingredient names.**
recipe_ingredients	**recipe_id + ingredient_id**	**Connects recipes with ingredients and stores quantities.**
recipe_steps	**step_id**	**Stores cooking instructions for each recipe.**

3. Table Relationships

The tables are connected using primary keys and foreign keys.

- One user can create multiple recipes.
- One category can contain multiple recipes.
- One recipe can have several cooking steps.
- One recipe can use multiple ingredients.
- One ingredient can be used in multiple recipes.

The recipe_ingredients table connects recipes and ingredients because the same ingredient can be used in different recipes.

These relationships help organize the database and reduce repeated information.

The `recipe_ingredients` table connects recipes and ingredients.

These relationships help keep the database organized and avoid repeating information.

4. Connection to Our Five Use Cases

Our group selected five main use cases in Deliverable 2.

Use Case	Tables Used	Explanation
UC-01: Create Recipe Entry	recipes, recipe_steps, recipe_ingredients	Adds new recipes with instructions and ingredients.
UC-02: Edit Recipe Entry	recipes, recipe_steps, recipe_ingredients	Allows recipe information to be updated.
UC-03: Search Recipe Entry	recipes, ingredients, recipe_ingredients	Allows recipes to be found by name or ingredient.
UC-04: Create Ingredient Entry	ingredients	Adds new ingredients to the database.
UC-05: Edit Ingredient Entry	ingredients	Updates existing ingredient information.

These tables provide the information needed to support the five use cases. The actual database actions can be performed using SQL.

5. Sample Data

Our `DATA.sql` file includes example information for testing the database.

There are four sample recipes:

- Garlic Bread
- Chicken Alfredo
- Chocolate Lava Cake
- Caesar Salad

The sample data also includes 12 ingredients, four users, four recipe categories, and cooking instructions.

6. Docker and Database Setup

Our group prepared a Docker Compose file using PostgreSQL 15.

The configuration includes:

- Database: projectdb
- Username: student
- Container: my_project_db

Docker Compose is used to start and stop the database container.

Once the SQL files are compatible with PostgreSQL and the database has started, we can connect using:

```
docker compose exec db psql -U student -d projectdb
```

We can then check the tables using:

```
SELECT * FROM recipes;

SELECT * FROM ingredients;
```

7. Conclusion

For Deliverable 3, our group designed a database structure based on the five use cases from Deliverable 2. We created six related tables and prepared sample data for recipes, ingredients, and cooking steps.

We also prepared a Docker Compose configuration to run PostgreSQL in a container. The original SQL scripts use MySQL syntax and will need to be adjusted before they can be used with PostgreSQL.